

How Much Does Qualifying Position Determine Race Outcomes in Formula 1?

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1. Introduction and Background

The pinnacle of motorsport is generally agreed to be the sport of Formula 1 (F1). It is a globally syndicated sport where drivers compete for the best possible time with various engineering advances and the designer of the car, against opponents with the same objectives and aims. Traditionally, in most sports, when players compete against each other, they all start at the same time. This is not true for F1. The F1 sport features a qualifying session that is done on the same day as the race, but depends on a lot of various factors. At the end of the qualifying, whatever driver is fastest, at least in most conditions, gets the starting position the furthest ahead on the qualifying grid. Starting the race at the very front of the grid on the very front of the grid confers and translates into a lot of race advantages.

There is a lot of popular belief and widespread analysis done by fans and commentators that there is a very close relationship and dependency on the qualifying performance and the overall finishing performance of the driver at the end of the race. The beliefs and concerns are all understandable on the industry leading designs and the overall winning performances that has been observed in the F1 industry in the modern day. The concept of dirty air and the close to non-existent ability to perform an overtaking maneuver due to the aerodynamic and engineering challenges that are present in front of that opposing driver is well evident in the industry. However, regardless of the observations and beliefs, the complete and precise relationship between the various starting positions and the drivers at the end of a race has not yet been thoroughly and successfully observed and measured as we are currently in the modern hybrid era of racing (2014-2024).

Our core findings: qualified position is a strong structural determinant of race outcomes, particularly winning, but its effect varies substantially across circuits and can be partially disrupted by Safety Car events. Generally, its importance has remained high over time, without demonstrating a consistently one-directional trend.

2. Research Questions

This project investigates the following research questions:

Main Research Question:

- How much does qualifying position determine race outcomes in Formula 1?

Sub-questions:

- RQ1: How does qualifying position affect the probability of winning a race?

- RQ2: How do Safety Car or Red Flag events affect position changes during a Formula 1 race?
- RQ3: Does the effect of qualifying position vary across different circuits?
- RQ4: Has the importance of qualifying position changed across recent Formula 1 seasons?

There are a few reasons why these types of questions are appropriate for analytical purposes. First, race data, which are structured and available in the public domain, can be used to answer these questions in a measurable way. Qualifying and finishing positions for each driver in every race are publicly available. Second, the questions are suitable for multi-dimensional analysis of the questions, in terms of overall, circuit, race, or seasonal breakdowns. Finally, the questions are likely to be connected to outcome determinants — podiums, position changes, and so forth. The data-oriented nature of the questions helps the authors of this analysis to move beyond the assumptions of anecdotal data and assess whether the qualifying importance of drivers meets the criteria of evidence through data.

3. Data Collection

3.1. Data Sources

We collected data from two publicly available, non-commercial APIs:

- **Jolpica F1 API (Ergast-compatible):** Provides structured historical F1 data including qualifying results (grid positions) and race results (finishing positions) for all seasons. We used this for RQ1, RQ3, and RQ4, targeting seasons 2014–2024 to capture the modern hybrid era with relatively consistent regulations.
- **OpenF1 API:** Provides real-time and historical event-level race data, including intra-race position telemetry and race control events such as Safety Car and Red Flag deployments. We used this for RQ2, limited to seasons 2023–2024 where event-level data was consistently available.

3.2. Collection Process and Challenges

For the Jolpica data, an initial attempt using a single large limit parameter returned incomplete results — the API silently truncated responses without error. After identifying this issue, we rewrote the collection script (`data_collection_jolpicafl.py`) to use offset-based pagination, iterating through all available pages for each season. HTTP 429 (Too Many Requests) rate-limiting errors required adding retry logic and request delays. To improve stability, collection was split into two time ranges (2014–2019 and 2020–2024), saved as separate CSV files, and later merged using `merge_jolpicafl_parts.py` via `pd.concat()`.

For the OpenF1 data, some session keys had no associated data (HTTP 404), handled by returning empty lists and skipping those sessions gracefully. Raw session-level CSV files were excluded from the GitHub repository using `.gitignore`, with only processed summary files tracked.

The final merged Jolpica dataset was verified using `df.groupby('season').size()`, confirming realistic row counts in the hundreds per season — consistent with approximately 20 races × 20 drivers per season.

4. Data Description

4.1. Dataset Scope and Size

The project uses two processed datasets:

- **f1_data_full_2014_2024.csv (Jolpica):** Covers 11 Formula 1 seasons (2014–2024). Each row represents one driver-race observation. The dataset contains hundreds of rows per season (approximately 400 per season: ~ 20 races $\times$ ~ 20 drivers). Used for RQ1, RQ3, and RQ4.
- **OpenF1 processed summaries:** Covers 2023–2024 only (2 seasons).
openf1_position_summary.csv contains driver-level position movement data;
openf1_race_control_summary.csv contains session-level race interruption counts. These were merged on session_key to produce a combined dataset of 918 driver-race observations for RQ2.

4.2. Key Variables

The table below summarizes the key variables used across all four research questions:

Variable	Type	Description
season	Numerical (int)	F1 season year (2014–2024)
grid	Numerical (int)	Driver's qualifying/starting position
finish	Numerical (int)	Driver's final race position
win	Binary (0/1)	1 if finish == 1, else 0
top3	Binary (0/1)	1 if finish $\leq$ 3, else 0
position_range_openf1	Numerical (int)	max_pos – min_pos per driver per race (OpenF1 only)
safety_car_events	Numerical (int)	Count of Safety Car deployments per race session
pole_win_rate	Numerical (float)	Proportion of races won from pole (season or circuit level)
avg winner grid	Numerical (float)	Average starting grid position of race winners
top3 win share	Numerical (float)	Proportion of winners who started in top 3

Table 1: Key variables used in analysis.

4.3. Dataset Characteristics

The Jolpica dataset spans the modern hybrid era, chosen to limit the influence of major regulatory changes and promote season comparability. The OpenF1 dataset is more limited in temporal scope but provides event-level granularity unavailable in Jolpica. Qualifying results and race results were merged using common keys (season, round, race name, driver). Approximately

1–2% of observations had missing or inconsistent data and were excluded during cleaning. No imputation was applied; removed observations are considered negligible given the dataset size.

5. Analysis

5.1. RQ1 — Qualifying Position and Win Probability

Method

Using `f1_data_full_2014_2024.csv`, we computed grid-level summary statistics: win rate (proportion of drivers who won from each starting position), top-3 rate (proportion finishing in the top 3), and sample size per grid position. A binary win variable was created (`win = 1` if `finish == 1`). Bar charts were used to visualize both outcome measures by starting grid position.

Results and Visualizations

The results reveal a strong and clear relationship between starting position and race outcomes.

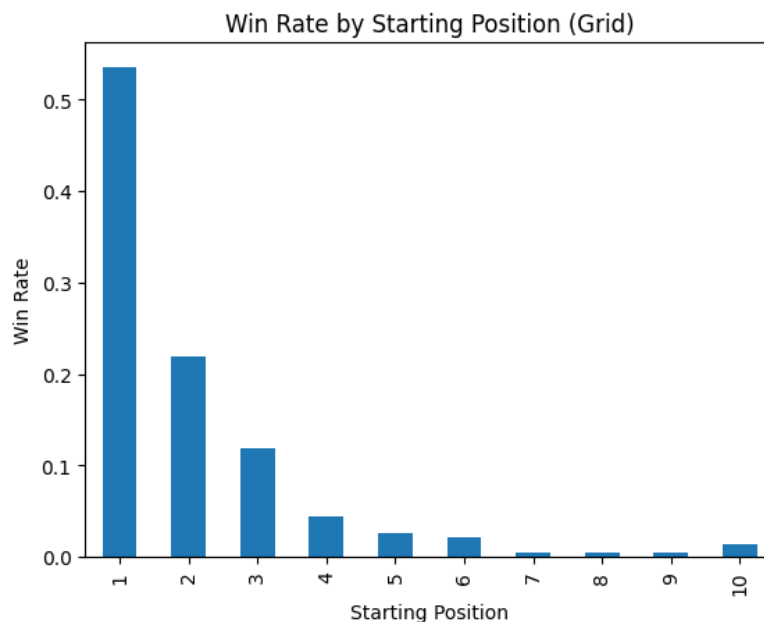


Figure 1: Win Rate by Starting Position. Pole position (grid 1) carries a dramatically higher win rate than any other position. Win probability drops sharply after the first few grid spots and becomes negligible beyond grid 6.

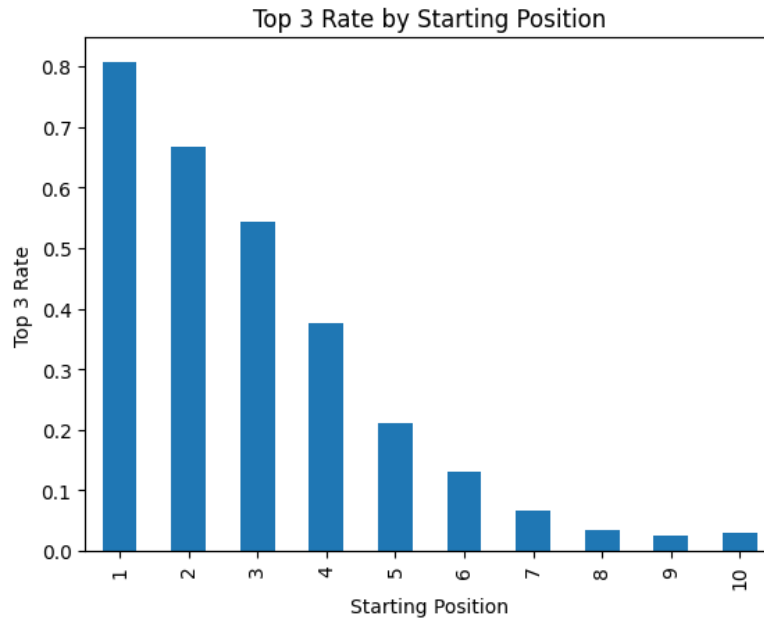


Figure 2: Top-3 Rate by Starting Position. Top-3 probability also declines with worse starting position, but more gradually than win probability — indicating that podium finishes are somewhat more accessible from lower starting positions than outright wins.

Interpretation

These findings confirm that qualifying position is a major structural determinant of race outcomes in Formula 1. The advantage of pole position is especially pronounced for winning: grid 1 produced wins at a rate roughly 2–3× higher than grid 2, and the probability becomes very small beyond grid 4. The more gradual decline in top-3 rate indicates that while winning is heavily front-loaded, podium finishes remain somewhat more open — consistent with the idea that 3 available podium spots spread the advantage slightly more broadly than 1 win.

5.2. RQ2 — Safety Car Events and Position Variability

Method

Using the OpenF1 2023–2024 dataset, we merged `openf1_position_summary.csv` with `openf1_race_control_summary.csv` on `session_key`. The key metric was `position_range_openf1` (`max_position` – `min_position` per driver per race), used as a proxy for intra-race position variability. We grouped races by number of Safety Car events and computed average position variability per group. Observations with more than 4 Safety Car events were excluded to avoid instability from very small sample sizes ($n < 20$).

Results and Visualizations

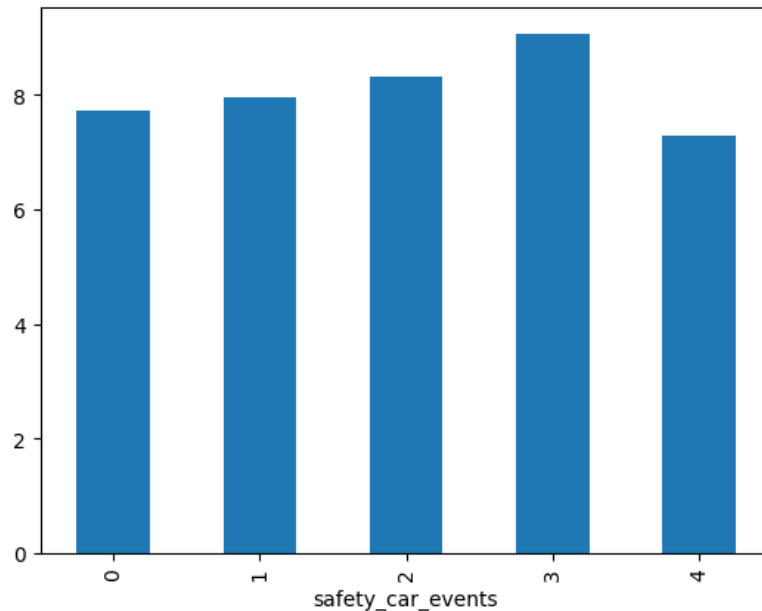


Figure 3: Average Position Variability (*position_range_openf1*) by Safety Car Event Count. Variability rises from 0 to 3 Safety Cars, peaking at 9.08, then drops at 4 Safety Cars (7.28) — suggesting a non-linear, inverted U-shaped relationship.

Average position variability by Safety Car count: 0 → 7.73, 1 → 7.95, 2 → 8.31, 3 → 9.08 (peak), 4 → 7.28. Note: group sizes are very unequal (0 SC: n=400; 1 SC: n=20; 2 SC: n=200; 3 SC: n=79; 4 SC: n=99).

Interpretation

Results suggest a non-linear relationship between Safety Car frequency and position variability. A moderate number of Safety Car events (2–3) creates strategic complexity — varied pit stop timing, different restart positioning — that maximizes reshuffling. When interruptions become excessive, the race is overly neutralized, reducing effective green-flag racing time and limiting additional position changes. This result should be interpreted as an association, not a causal claim. The small n=20 for the 1 Safety Car group means its mean is less reliable than the other groups.

5.3. RQ3 — Circuit-Level Variation in Qualifying Importance

Method

Using *f1_data_full_2014_2024.csv* filtered to winners only, we constructed three circuit-level summary indicators using *race_name* as a circuit proxy: (1) *avg_winner_grid* — average starting position of race winners, (2) *pole_win_rate* — proportion of races won from pole position, (3) *top3_win_share* — proportion of winners who started in the top 3. Only circuits with at least 8 race observations were included to avoid unstable estimates.

Results and Visualizations

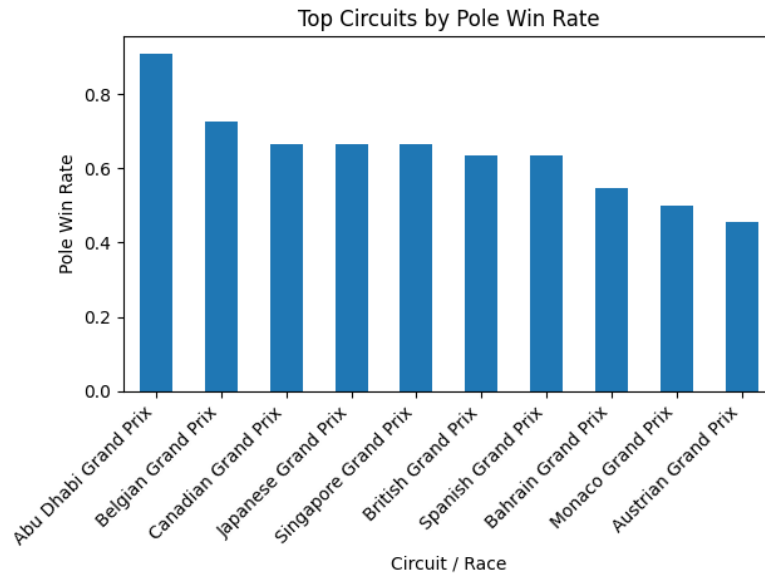


Figure 4: Top Circuits by Pole Win Rate. Abu Dhabi (0.909) and Belgian (0.727) Grand Prix show the highest pole-to-win conversion, indicating strong dependence on qualifying position.

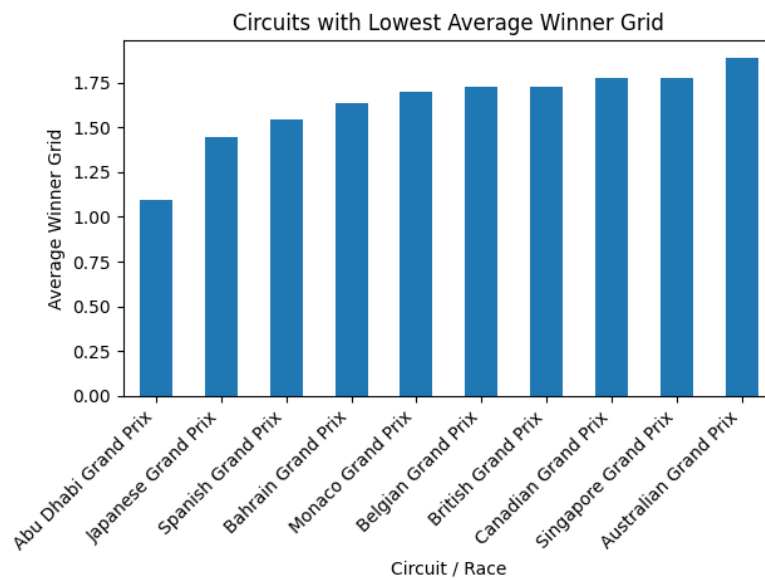


Figure 5: Circuits with Lowest Average Winner Grid. Abu Dhabi (1.09) and Japanese (1.45) Grand Prix show winners almost always starting from the very front of the grid.

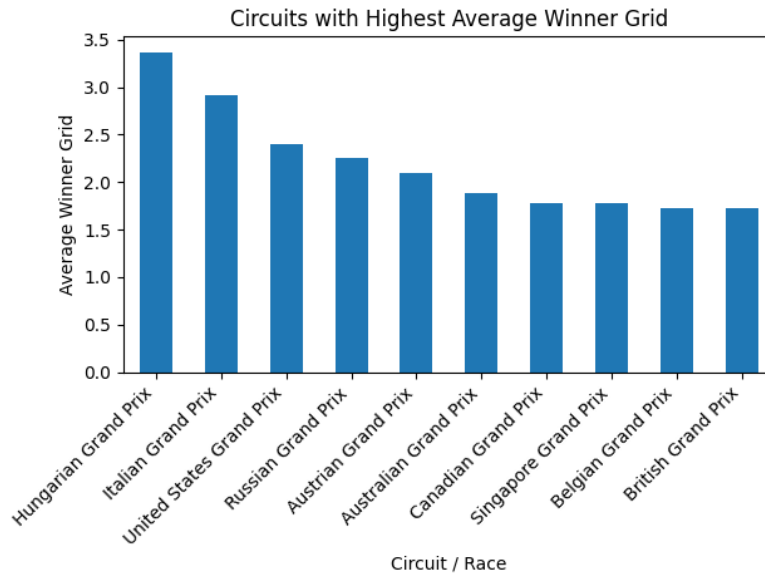


Figure 6: Circuits with Highest Average Winner Grid. Hungarian (3.40) and Italian (2.91) Grand Prix show winners starting further back on average, indicating more opportunity for non-front starters to win.

Interpretation

Results confirm that qualifying importance is not uniform across circuits. Circuits like Abu Dhabi and Monaco are strongly front-loaded — their track characteristics (narrow layout, limited overtaking zones, clean air advantage) heavily reward starting position. Circuits like Monza (Italian GP) and Hungaroring show more variance in winner starting positions, suggesting that race strategy, tire management, and Safety Car timing play larger roles there. Note that `race_name` is used as a circuit proxy and does not directly encode physical track characteristics such as DRS zone count, circuit length, or overtaking difficulty ratings.

5.4. RQ4 — Temporal Trends Across Seasons

Method

Using `f1_data_full_2014_2024.csv`, we computed three season-level indicators: `pole_win_rate`, `top3_win_share`, and `avg_winner_grid`. We visualized these as time series (2014–2024) and compared averages across two broader periods: 2014–2019 (pre-regulation change) and 2020–2024 (post-regulation change era).

Results and Visualizations

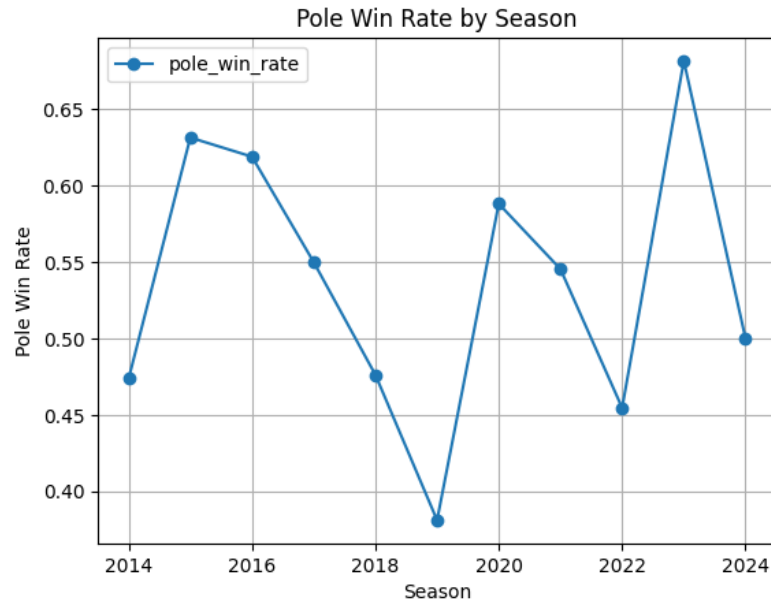


Figure 7: Pole Win Rate by Season. No consistent directional trend; notable spike in 2023 (Red Bull dominance), dip in 2019.

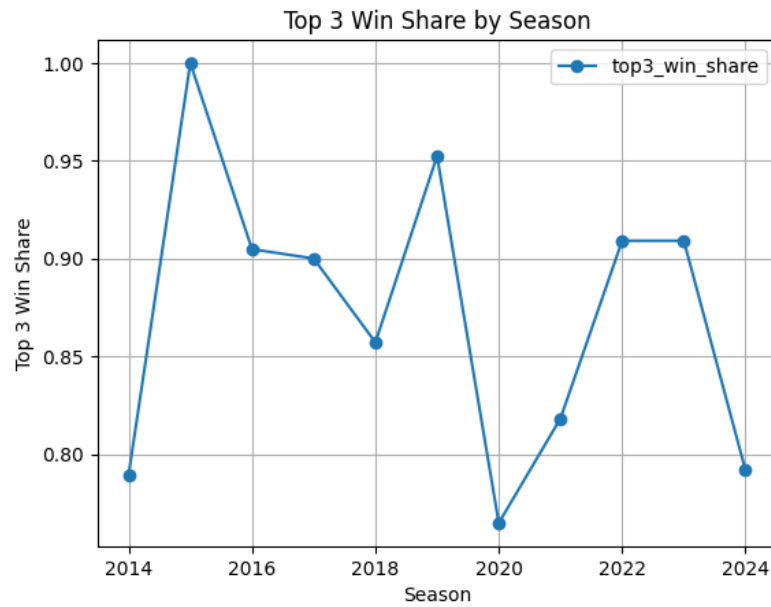


Figure 8: Top-3 Win Share by Season. Fluctuates without a clear trend; 2015 peak (1.00) and 2020 dip (~0.77) reflect season-specific competitive dynamics.

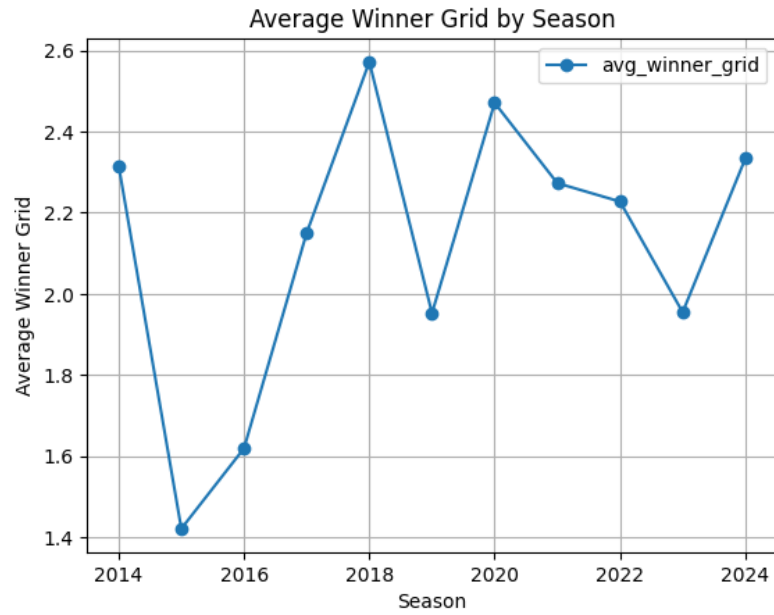


Figure 9: Average Winner Grid by Season. 2018 and 2020 show winners starting further back; 2015 shows the most front-loaded winners.

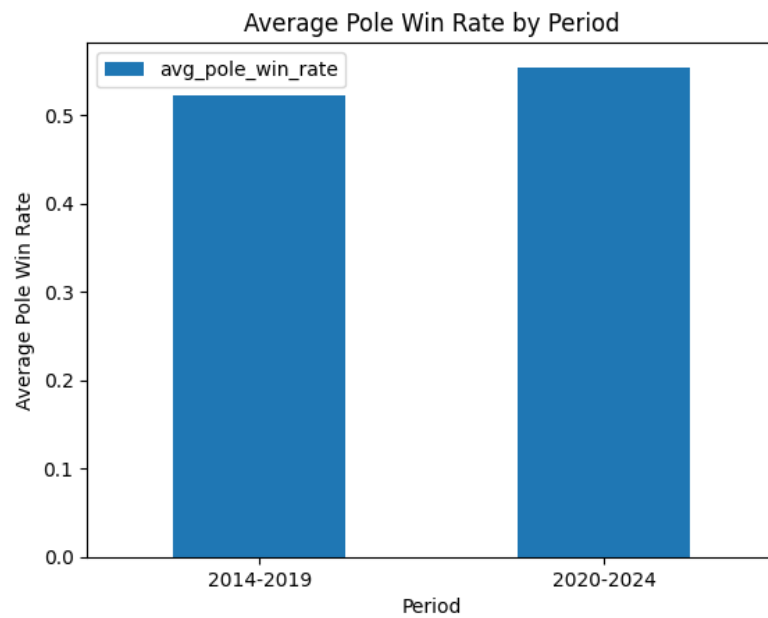


Figure 10: Average Pole Win Rate by Period. Slight increase from 0.522 (2014–2019) to 0.554 (2020–2024).

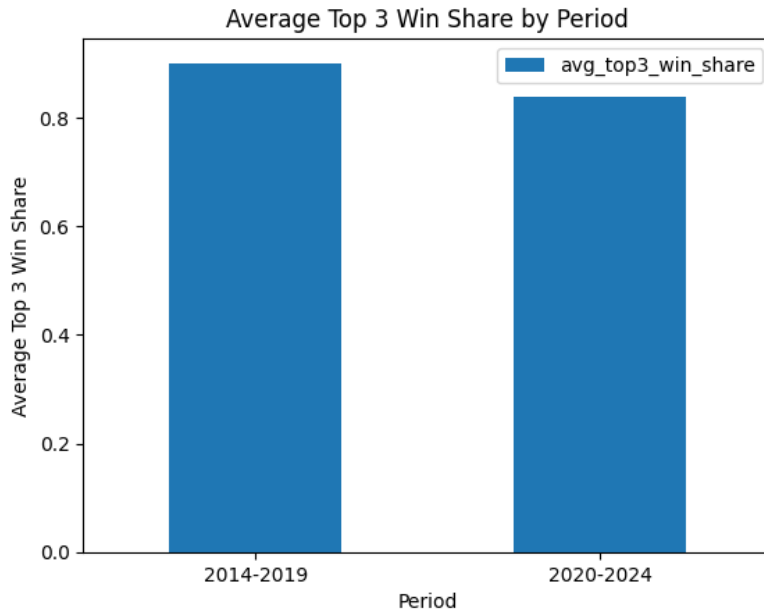


Figure 11: Average Top-3 Win Share by Period. Decreased from 0.901 (2014–2019) to 0.839 (2020–2024), suggesting slightly more winners from outside top 3 in recent years.

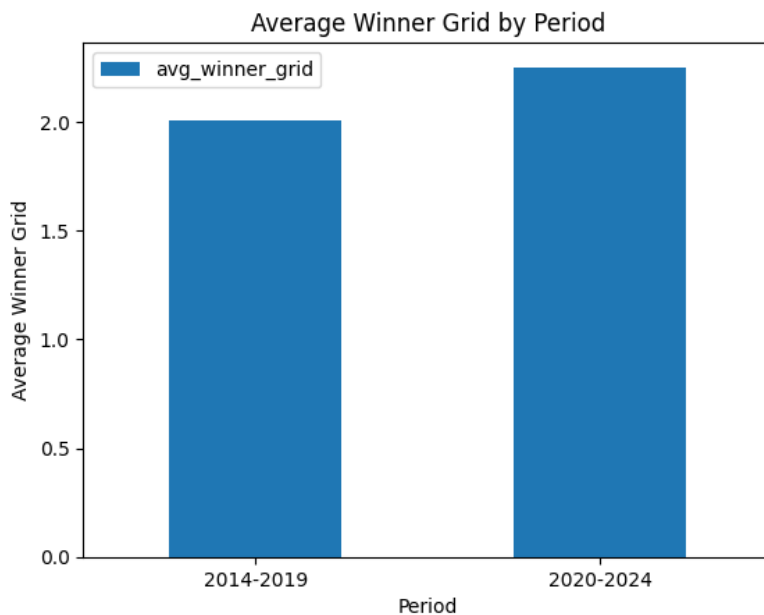


Figure 12: Average Winner Grid by Period. Increased from 2.005 (2014–2019) to 2.252 (2020–2024), indicating winners started slightly further back on average in the later period.

Interpretation

The three period-level indicators move in different directions: pole win rate increased slightly, but top-3 win share decreased and average winner grid increased. These contradictory signals mean no clean claim of 'qualifying has become more (or less) important' is warranted. Year-to-year variation is clearly present — 2023's high pole win rate reflects Red Bull's historically dominant season; 2020's lower top-3 share reflects a chaotic COVID-shortened season with

many unusual results. These factors highlight why aggregate trends are difficult to interpret without accounting for team dominance cycles and regulatory changes.

6. Conclusions and Reflection

6.1. Summary of Key Findings

The four research questions together build one coherent narrative about qualifying position in Formula 1:

- **RQ1 — Baseline structure:** Qualifying position is a strong and systematic structural determinant of winning. Pole position carries a commanding advantage; win probability drops sharply thereafter.
- **RQ2 — Disruption by race events:** Moderate Safety Car activity is associated with the greatest intra-race position variability. Excessive interruptions reduce effective racing time and may limit further reshuffling.
- **RQ3 — Circuit-level heterogeneity:** Qualifying importance varies significantly across circuits. Some tracks (Abu Dhabi, Monaco) strongly reward front-row starts; others (Hungarian, Italian GP) allow more variance in winner starting positions.
- **RQ4 — Temporal stability:** Qualifying importance has remained consistently high across seasons, but shows no clear one-directional trend between 2014–2019 and 2020–2024.

In summary: Formula 1 race outcomes are strongly structured by qualifying position — but that structure is not equally strong across all races, it can be partially disrupted by race interruptions, and it has not shifted systematically across recent seasons.

6.2. Implications

These results show that some aspects of Formula 1 can be laid bare as a competitive sport. Firstly, when a significant qualifying outcome relationship exists, it raises pertinent issues around competitive equity. Simply, if a starting position is a strong predictor of victory, then for fans looking to see an exciting spectacle, many outcomes could be decided long before the Grande Prix. Second, the circuit-level estimates show not all races are of the same competitive equity. It is shown that some circuits are more conducive to racing. It is shown that the chance of seeing an overtaking maneuver on an average of one lap = 3, may be a lot more than on a circuit where overtaking maneuvers are a rarity. Third, it may also alter the way race formats are assessed (i.e. sprint races or race formats that use a reverse grid) as other race day outcomes that are random or more chance driven may Reverse the order of qualifying.

6.3. Limitations

- **API data incompleteness:** Jolpica's silent pagination truncation required detection and a full script rewrite. OpenF1 event-level data was only reliably available for 2023–2024, preventing integration with the full historical dataset.
- **Observational design:** All four analyses identify associations, not causal effects. Car performance, team strategy, tire compounds, weather, and driver skill are not controlled.

- **Proxy limitations:** `position_range_openfl` measures total range of position movement but not direction, timing, or strategic significance. `race_name` is used as a circuit proxy but does not encode physical track characteristics.
- **Sample size instability in RQ2:** The 1 Safety Car group (n=20) produces a less reliable mean than other groups. Interpretations involving that group should be treated cautiously.
- **Missing confounders:** Team dominance cycles (e.g., Red Bull 2023, Mercedes 2014–2016) strongly influence season-level metrics in RQ4. Interpreting temporal trends without controlling for team dominance may be misleading.

6.4. Reflection

This project taught us a lot of skills that we can use in the real world. We learned how to work with APIs, dealing with issues like pagination, rate-limits, and partial data. We also worked on making our GitHub repo be more organized and user-friendly by putting our collections, scripts, raw data, and notebooks on separate folders. We can use our repo structures again in the future. When we worked on our visual analytics, Python in Jupyter, we learned how to make indicators from our research questions, and appropriately use and interpret data visualizations.

We also learned a couple of things on the conceptual side. We learned the difference between an association vs. causation and how to figure this out from just observational data. We learned to check how many samples there actually are in order to make sense of grouped means. The RQ2 section in our project works to explain this. Breaking the original question down by event, of each type, in each respective series, and season works to support the statement that one single finding of an aggregate type of question is actually insufficient.

7. CRediT Author Statement

- Jun-Hyeok Lee: Data curation, API collection scripting (Jolpica and OpenF1), pagination and rate-limit handling, data merging and validation.
- Jihee Lee: Exploratory data analysis, visualization design, descriptive statistics, RQ1 and RQ3 analysis.
- Jae Yang: Statistical analysis, regression modeling, RQ2 and RQ4 analysis, results interpretation.
- Dong Hun Yoo: Project coordination, literature review, presentation design, final report writing and editing.

All members collaborated on research design, discussion of results, and final editing.

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